**MA3521 — In-Class Worksheet Tuesday/Wednesday**

Name:

Do not solve explicitly for $y$.

1. Linear $\quad ty' + 2y = e^t \qquad y(1) = 0$

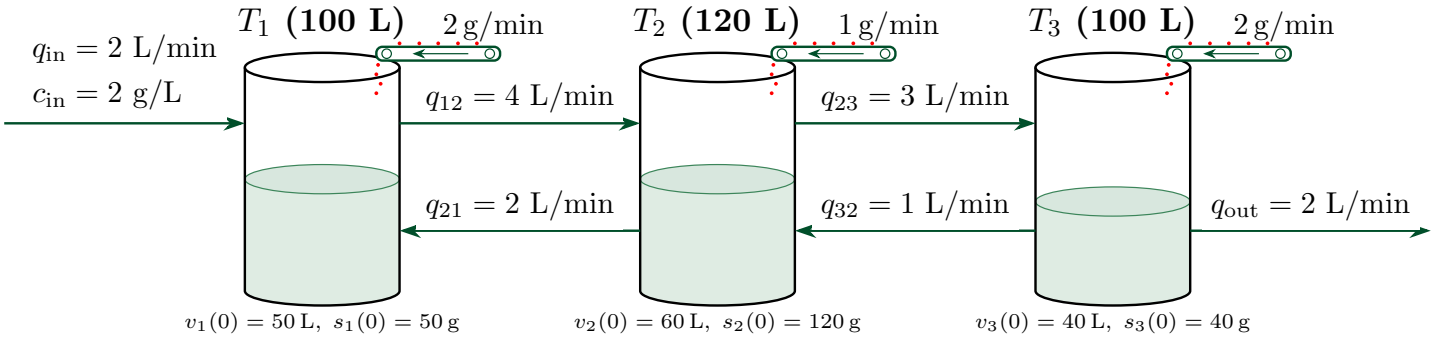
   Intermediate step

   Answer

2. Separable $\quad y' = 2t\cos(t^2)(1 + y^2) \qquad y(0) = 0$

   Intermediate step

   Answer




Well-mixed tanks, $t$ in min, $\mathbf{v}(t)$ in L, and salt $\mathbf{s}(t)$ in g. Dry Salt added at shown rates. Set up $\mathbf{s}' = \boldsymbol{A}\mathbf{s} + \mathbf{b}$.

1. $\mathbf{v}(t) =$ [ ] L $\quad \mathbf{s}(0) =$ [ ] g

2. $\boldsymbol{A} =$ [ ] $\text{min}^{-1}$ $\quad \mathbf{b} =$ [ ] g/min

3. A suitable UC form is $\mathbf{s}_p(t) =$ [ ]

4. To determine the UC coefficients you would row reduce

$\boldsymbol{M}_{\text{UC}} =$ [ ]